

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

If $u-3 = \frac{6}{t-2}$, what is t in terms of u ?

Answer

- A. $t = \frac{1}{u}$
- B. $t = \frac{2u+9}{u}$
- C. $t = \frac{1}{u-3}$
- D. $t = \frac{2u}{u-3}$

Correct Answer: D

Rationale

Choice D is correct. Multiplying both sides of the given equation by $t-2$ yields $(t-2)(u-3) = 6$. Dividing both sides of this equation by $u-3$ yields $t-2 = \frac{6}{u-3}$. Adding 2 to both sides of this equation yields $t = \frac{6}{u-3} + 2$, which can be rewritten as $t = \frac{6}{u-3} + \frac{2(u-3)}{u-3}$. Since the fractions on the right-hand side of this equation have a common denominator, adding the fractions yields $t = \frac{6+2(u-3)}{u-3}$. Applying the distributive property to the numerator on the right-hand side of this equation yields $t = \frac{6+2u-6}{u-3}$, which is equivalent to $t = \frac{2u}{u-3}$.

Choices A, B, and C are incorrect and may result from various misconceptions or miscalculations.